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THE NEXUS BETWEEN CLIMATE POLICY UNCERTAINTY AND LOAD CAPACITY FACTOR IN CHINA WITH THE MODERATING ROLE OF FINANCIAL DEVELOPMENT

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Abstract

The current study explores the role of financial development and climate policy uncertainty on environmental quality in China, one of the countries with the highest levels of environmental pollution, during the period 1990-2021. This study uses the load capacity factor to indicate environmental quality, while financial development and climate policy uncertainty, energy consumption and economic growth, are considered as explanatory variables. Two different models are constructed to explore the mediating role of financial development and climate policy uncertainty on environmental quality. Analyses are performed using the A-ARDL approach and the FMOLS long-term estimator, and the DOLS analysis is utilized for robustness. The findings obtained using different approaches support each other. According to the empirical results, financial development and climate policy uncertainty reduce environmental quality in China, but improve it when the interaction variable is integrated into the model. This finding implies that the association between financial development and climate policy uncertainty can play a balancing role in ensuring environmental sustainability.

Keywords: Environmental quality, Financial development, Climate change, Load capacity factor, Augmented ARDL.

JEL Codes: Q57, E44, Q54, P18

1. Introduction

Since 1850, global temperatures have shown an average increase of 0.06°C every decade (NOAA, 2023). Although this change in average global temperatures seems like a small increase, it significantly affects the heat energy of water resources, soil, and the atmosphere (Lindsey & Dahlman, 2025). The effects of human-induced (Banelienė & Strazdas, 2025) climate change (CC), most of which are irreversible, reached new peak values in

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2024 and are projected to continue increasing in the next years. In 2024, the year with the highest recorded temperatures in the past 175 years, CC accelerated migration and caused significant damage to forests and biodiversity (WMO, 2025).

The CC increases drought and water stress, causes diseases, leads to the loss of ice caps and consequently sea level rise, causing soil erosion (Nguyen & Ngo, 2022; Adebayo et al., 2024a; Adebayo et al., 2024b). In addition, it disrupts underwater ecosystems and leads to ocean acidification, threatening marine life (NAOO, 2025). Furthermore, environmental deterioration is being intensified by CC, which results in severe natural disasters (IOM, 2011).

Different indicators of environmental degradation address different aspects of environmental health. The ecological footprint (EF) represents the biologically efficient land and sea areas stipulated by a population to sustain its resource use and absorb the waste it produces (especially carbon emissions). In this context, the EF is calculated by combining multiple land uses, including agricultural land, pastures, fishing grounds, built-up areas, forest resources, and ecological CO₂ storage capacity (Global Footprint Network (GFN), 2025). Therefore, the EF is a more extensive measure than CO₂ (Nathaniel et al., 2021). However, the EF considers the demand for natural resources but ignores biocapacity. The Load Capacity Factor (LCF), on the other hand, is a more holistic measure of environmental degradation as it also considers biocapacity (BC) (Pata & Samour, 2022). Therefore, we focus on LCF as an indicator of environmental quality (EQ). China has a special significance at the global level in terms of environmental pollution.

The LCF, an indicator of EQ, was 0.508 in China in 1990 and 0.199 in 2021. Accordingly, there has been a significant decline in China's EQ for the last few decades (GFN, 2025). China has strong potential to combat CC on a global scale. China invested in clean energy technologies to achieve its carbon neutrality goal and combat environmental degradation and CC. Nonetheless, environmental policies of China are inadequate and unclear (Climate Action Tracker, 2025). China needs effective and holistic policies to mitigate, or even eliminate, the undesirable impacts of climate policy uncertainties (CPU) on EQ. However, previous studies have been far from providing policymakers with clear conclusions. Therefore, this research examines the moderating effect of financial development (FD) in reducing or completely eliminating the undesirable consequences of CPU on EQ in China, and offers holistic policy recommendations for China, also considering the consequences of EG and energy utilization.

Undesirable effects of CPU on EQ have been frequently reported in the literature (Dong & Zheng, 2025). Uncertainties can make it difficult for firms to take decisions about carbon pricing and the environment, thus hindering them from making environmentally friendly investments. This can lead companies to engage in dirty production activities. Hence, environmental

policies can give way to short-term goals, and the carbon neutrality target may be abandoned (Sun et al., 2025). Previous studies showed that FD can serve as a mediating mechanism affecting EQ during periods of uncertainty (Chu et al., 2023).

FD has the capacity to improve EQ (Durmus et al., 2025). FD can contribute to improving EQ by providing funds and diversifying green financing alternatives (UNFCCC, 2025). Furthermore, FD can enhance EQ by supporting the transition from fossil fuel-based energy use to low-carbon energy sources (Horky & Fidrmuc, 2024). In addition, FD can contribute to sustainability by mitigating the negative consequences of CPU on the EQ (Kong et al., 2025). Three different channels can be effective in this context. The first is the investment and innovation channel. Deep financial systems in this channel can increase green investments. Increasing financial depth stimulate green energy projects and green R&D activities. Long-term financing can be provided for environmentally friendly technologies. In particular, banking and capital markets can continue to investments prioritizing EQ during periods of uncertainty, limiting the investment contraction caused by the CPU (Sun et al., 2025). The second channel can be defined as the risk sharing and resilience channel. During periods of CPU, deepening financial systems can minimize uncertainty-related risks through effective credit mechanisms and a variety of financial instruments (e.g., green bonds). Therefore, during periods of uncertainty in climate policy, environmental protection can be ensured by distributing risks within the financial system without reducing environmental investments (Subhani et al., 2025). The third channel is the resource allocation and knowledge generation channel. It is stated that FD can increase the capacity for knowledge generation and project screening in this channel. This allows financial institutions and organizations to effectively distinguish between emission-increasing projects and green projects. This implies that EQ can still be financed with available funds despite the decline in total investment during CPU periods (Chu et al., 2023). However, the association between CC and finance can harm to financial inclusion, negatively affect resource allocation, and deteriorate the financial system's functioning (Nur et al., 2023). Hence, the consequences of CC and the interaction between climate and finance are uncertain. This implies that more research is required in this area. Previous studies overlooked the mediating effect of FD on the association between CPU and EQ. This research gap forms the main motivation of the current study.

FD can act as a balancing mechanism in the relationship between climate policy uncertainty and EQ through various channels. First, the existence of deepened financial markets can increase investments prioritizing EQ by increasing the diversity of financial instruments and minimizing financing constraints (Gharbi et al., 2025). Additionally, high levels of FD can lower firms' risk perceptions at the micro level. This

increases firms' ability to protect themselves against potential market risks and significantly reduces their perception of uncertainty. Third, financial markets increase information efficiency. Providing complete information increases the likelihood of resources being used for more environmentally sensitive activities. Thus, resource efficiency increases, minimizing environmental stress (Awewomom et al., 2024). Fourth, advanced FD can lead to stronger regional and national public sector investments in environmental protection. This can be followed by private sector investments in environmental protection (Tamazian & Rao, 2010). Conversely, an underdeveloped financial system can lead to climate policy uncertainty and the postponement of investments. This procrastination behavior leads to a failure to reduce the use of non-renewable energy sources and an increase in dependence on polluting sources. For these reasons, the interaction between FD and climate policy uncertainty can lead to significant environmental consequences. Thus, it is increasingly expected that these variables will play an important role in determining environmental policies through relevant mechanisms. There are also some studies, albeit fewer, that argue that the positive regulatory effect of FD is not always valid (Rahman et al., 2025). In this context, with FD and a favorable credit environment, investments in activities that increase carbon dioxide emissions can increase. For example, increased investments in things like houses and cars increase energy consumption, and these situations often lead to this increase in environmental pollution (Lv & Li, 2021).

The literature generally associates the EG-EQ nexus with the EKC (Oladunni et al., 2024). Previous studies emphasized that the EG-EQ nexus follow an inverted U-shaped linkage, thus supporting the validity of the EKC hypothesis. Accordingly, environmental degradation increases as income surges in early levels of EG, but after a particular level, income improves EQ (Şimşek et al., 2025; Özbek & Oğul, 2022). In this process, EQ shows an initial decrease due to industrial production and resource-intensive manufacturing activities. In the second stage, at higher income levels, countries seek out environmentally friendly technologies, and as they adopt more environmentally friendly technologies, implement stricter regulations, and reduce the weight of the manufacturing industry sector in output, environmental pollution decreases. The fact that the EG-EQ nexus facilitates an inverted U-shaped linkage is consistent with the LCC hypothesis. This hypothesis, like the EKC, suggests that LCF decreases in parallel with a decrease in EQ during the initial stages of income level, but increases during periods when EQ begins to improve. LCC is a hypothesis that allows for the simultaneous evaluation of EQ and LCF (Güler et al., 2025; Ridwan et al., 2024).

Global declines in EQ and climate change pose serious risks to sustainable development plans, which have become a global goal. Within this context, a new perspective on environmental sustainability, differing from

the existing literature, has become necessary. In this regard, a thorough reassessment of the relationships between EG, energy consumption, and FD has emerged. China, an emerging market economy undergoing rapid industrialization and financial deepening, is a central country in these discussions. During the period of globalization (1990-2021), increasing energy demand, growing FD, and high EG rates in China have been among the factors increasing environmental stress.

Existing studies in the literature cover a wide range of research on the relationship between finance, EG, energy, and the environment. However, a significant portion of these studies examine the relevant factors within the framework of singular and linear relationships. This leads to insufficient consideration of the interactions between variables, and especially the role of policy uncertainty. On the other hand, despite the increased interest in recent years due to the serious impact of climate change, there are a limited number of studies on the environmental effects of climate policy uncertainty. In this context, it is observed that the interactions between policy uncertainty and other macroeconomic variables are excluded. In the literature, environmental indicators with high usage frequency, such as CO₂, do not address environmental stress in a multidimensional way and inadequately reflect the balance between ecological capacity and human activities. This situation reveals the contribution of the LCF, which is a more holistic environmental indicator.

The present study offers three specific contributions to academic literature. (i) Previous studies ignored the moderating effect of CPU on the linkage between EQ and FD. Based on the authors' knowledge, this study is the first to concentrate on the moderating effect of FD and CPU on EQ. This study facilitates the literature by concentrating on the impact of FD, which have been overlooked in the literature, on the consequences of CPU-EQ nexus. The examination of this relationship reveals important implications for policymakers in terms of limiting the negative consequences of CPU on EQ and even transforming them into positive effects. (ii) Second, this study delivers policymakers with an integrated perspective by exploring the economic, social, and political determinants of EQ. This research offers insights into the multidimensional nature of environmental dynamics. It is hoped that this can provide a holistic analytical framework that enables policymakers to develop more coordinated and evidence-based environmental policies. (iii) Finally, to the authors' knowledge, this is the first study to explore the consequences of CPU on biocapacity. This study, conducted on China, a major global polluter, contributes significantly to addressing a gap in the literature and informing policymakers.

In the second part of the study, previous studies are summarized, in the third part the data methods are explained, in the fourth part the findings are summarized, in the fifth part the results are explained and discussed, and finally, in the sixth part policy recommendations are presented.

2. Literature

This section will examine the relationship between LCF and FD, CPU, energy consumption (EC), and EG separately. Specifically, the FD-LCF, CPU-LCF, EC-LCF, and EG-LCF relationships are discussed. The study also indirectly refers to research on the environmental impacts of the interaction variable consisting of FD and CPU variables, which constitutes the original contribution of this work and underlines its starting point. Finally, following the literature review, there is a literature gap section. This section highlights the research gap and emphasizes the main contribution of this study.

2.1. Financial Development and LCF

It is known that previous studies in literature on the interaction between FD and EQ reached different conclusions. Increased FD can, in a sense, pave the way for increased economic activity, thereby increasing energy demand. This can reduce EQ, especially in economies that largely depend on fossil fuel energy sources. Studies assessing the consequences of FD on LCF can be categorized into two main groups. While some studies show that FD reduces LCF and thus lowers EQ, others demonstrate that FD increases LCF and improves EQ.

Adebayo et al. (2025) determined that FD is an effective tool for increasing LCF in the US. The authors found that FD is not effective in improving EQ in the short run, but it increased EQ in the medium and long run by financing renewable energy investments. Nuta et al. (2025) reported that FD enhances EQ in EU nations for the 1996-2019 period using the two-step GMM approach. Ali et al. (2025) found that FD has an impact on LCF in France and supports SDGs, thus improving EQ. The authors stated that FD increases LCF by enabling the necessary financing for clean energy and promoting clean production technologies. Altıntaş et al. (2024) demonstrated that banking sector development, one of the primary measures of FD in Malaysia, supports EQ by increasing LCF. The authors stated that a strong and deep financial sector supports sustainability. Accordingly, encouraging the financing of environmentally friendly projects, supporting green finance initiatives, and ensuring the adoption of environmentally sound risk management principles for financial institutions can improve EQ by increasing LCF.

On the contrary, Bashir et al. (2022) argued that FD negatively affected EQ in RCEP countries during the period 1990-2019. Awosusi et al. (2024) used the ARDL boundary test to conclude that FD has a negative impact on LCF, thereby decreasing EQ, using annual data for Japan from 1980 to 2017. Similarly, Ridwan et al. (2024) reported that FD negatively impacts LCF in the US, and increased environmental degradation between 1990 and 2022. The authors stated that the LCC hypothesis holds for the US.

Güneysu (2023) investigated the validity of the LCC hypothesis for Türkiye and the effect of FD on LCF using FMOLS and DOLS analyses in the

period 1970-2018. The author showed that FD and LCF can interact in the long term, and that FD's effect on LCF is negative. Furthermore, the author stated that the LCC hypothesis holds for Türkiye, indicating an inverted U-shaped relationship between EQ and income. Coban et al. (2024) examined the level of achievement of Türkiye's SDGs using quantile and Granger causality analyses. The authors found that the consequences of FD on LCF is negative in most quantiles, but more so in the medium and high quantiles.

Javed et al. (2025) scrutinized the consequences of FD on LCF in G7 countries and the credibility of the LCC hypothesis, emphasizing that the determinants of LCF have been the subject of research in very few studies in the literature. The authors showed that FD reduces LCF in G7 countries, thus negatively affecting EQ. Furthermore, the authors stated that the LCC hypothesis holds for these countries. Wang et al. (2024a) concluded that FD and EG increased pollution by reducing LCF in BRICS countries between 1990 and 2018, thereby reducing EQ. A similar study and results have been reported by Li et al. (2024). The authors conducted CS-ARDL analysis for the same period and concluded that FD and EG had a disruptive effect on the ecological environment. Khan et al. (2025) analyzed the effect of FD on LCF while testing the LCC hypothesis for a data set covering the years 1990-2020, specifically for BRICST countries. Panel quantile regression findings verified the LCC hypothesis in these countries. Hence, the authors argued that FD negatively affects LCF, so reducing EQ. Durmus et al. (2025) based on the years 1991-2021 in a sample of newly industrialized countries, analysis using the Driscoll-Kraay and two-stage least squares methods revealed that FD negatively affects on LCF. Panel quantile regression findings showed that negative consequences of FD on LCF is statistically insignificant at high quantiles.

Wang et al. (2024b) used panel quantile analysis for Asian nations and found that FD increases LCF, thus positively affecting EQ. Furthermore, the authors demonstrated that the LCC hypothesis holds in these countries. Annor et al. (2024) determined that FD escalated LCF, and therefore improved EQ, in 47 countries.

2.2. Climate Policy Uncertainty and LCF

There is only one study (Pata, 2025) that explored the consequences of CC policy on LCF. Therefore, this section attempts to create a framework by exploring the consequences of CPU on other environmental pollution measures. Sun et al. (2023) determined that CPU reduce sustainability in the US. Sun et al. (2024) argued that increases in CPU undermine green innovation and thus environmental sustainability by reducing green incentives and making it more difficult for businesses to fulfill their environmental responsibilities. Pata (2025) demonstrated that CPU reduce the LCF using Fourier quantile causality analysis for China. The author stated that policymakers in China should consider the negative impacts of CPU on China's environmental sustainability.

Meo & Adebayo (2025) scrutinized the consequences of CPU on CO₂ in the US for the 1988-2022 period. The authors showed that CPU is the Granger cause of CO₂ in the US. Accordingly, CPU is primarily responsible for CO₂ emissions in this country. Huang et al. (2024) found that CPU in 47 BRI nations worsened EQ by increasing CO₂ and EF during the period 1998-2021. The authors emphasized the potential of measures to reduce CPU to improve EQ.

Liu et al. (2023) explored the short-term co-movement of CPU in the US. The authors emphasized that CPU moves in line with CO₂ in the short run, therefore an upsurge in CPU is linked with an upsurge in CO₂. Persakis et al. (2024) determined that CPU reduced EQ in 126 nations with different development levels during the period 1996-2021. The authors explained this result by stating that CPU encourages environmentally friendly behaviors in individuals. Yousfi et al. (2023) determined that CPU is the Granger cause of CO₂ in the US. The authors showed that increases in CPU can be associated with increases in CO₂, and that this relationship intensifies over periods of higher uncertainty, such as the Covid-19 pandemic.

On the contrary, Işık et al. (2025) showed that increasing CPU on a global scale reduces CO₂. The authors explain this result by stating that climate-related uncertainties reduce investments, thereby decreasing industrial activity and consequently emissions. Wang et al. (2024c), in their analysis of 1211 companies, showed that CPU increase the emissions caused by firms. The authors stated that strict environmental regulations and the promotion of social responsibility activities can limit the negative effects of CPU.

2.3. Energy Consumption and LCF

Reducing energy consumption is one of the key determinants that contributes to environmental protection (Roletto et al., 2024). Energy consumption (EC), which is generally based on the use of fossil fuels, results in a decrease in EQ. Replacing fossil fuels with clean energy can contribute to improving EQ (Ulussever et al., 2025). However, existing studies provide mixed results on the effect of energy consumption on EQ.

Pata & Balsalobre-Lorente (2022) scrutinized the consequences of energy consumption on LCF in Türkiye for the 1965-2017 period utilizing the ARDL approach. Pata & Balsalobre-Lorente (2022) demonstrated that energy consumption reduces LCF and therefore negatively affects EQ. Bergougui & Meziane (2025) found that EC has a positive effect on LCF in their analysis of Algeria using data from 1980 to 2023, but this result is not statistically significant.

Using data from Mexico for the period 1971-2018, Raihan et al. (2023) utilized Mexican data for the 1971-2018 period to scrutinize the consequences of fossil fuel and clean energy on LCF. The authors reported that fossil fuel energy consumption negatively affected LCF, thereby

reducing EQ; conversely, renewable energy consumption positively affected LCF, thereby increasing EQ. Hakkak et al. (2023) scrutinized the effect of clean energy on LCF using the ARDL technique for the period 1992-2018 in Russia, testing for the LCC hypothesis. Over the long term, the use of clean energy sources positively affects the LCF, and therefore it has been observed that clean energy improves EQ. The study also reports on the credibility of the LCC hypothesis. Uçar et al. (2025) investigated the impact of nuclear and hydroelectric EC on LCF in BRICS nations for the 1993-2022 period, also testing the LCC hypothesis.

Previous studies scrutinizing the consequences of clean energy on EQ lacked a consensus. Pata et al. (2023) stated that energy consumption is a fundamental factor reducing EQ, but the effect on EQ varies in line with the type of energy. Accordingly, clean energy sources can be an important policy tool in improving EQ. Yurtkuran & Güneysu (2023) revealed the potential of renewable energy to improve EQ in Türkiye using ARDL analyses.

2.4. Economic Growth and LCF

Previous studies scrutinizing the EG-LCF nexus have not reached a common conclusion. Yang et al. (2024) determined that in BRICS countries, EG during the 1990-2018 period reduced LCF, thus worsening EQ. Hammami et al. (2025) showed that in the top ten most innovative countries, EG reduced LCF in all quantiles during the period 1997-2020, thus negatively impacting environmental health. Raihan et al. (2023) examined long and short run correlations in Mexico for the 1971-2018 period utilizing the ARDL approach and determined that the consequences of EG on LCF is negative.

Duran & Saqib (2024) showed that using the CS-ARDL method in G20 countries between 1993-2021, the consequences of EG on LCF is negative in the short and long run, and therefore EQ is negatively impacted. Li et al. (2023) reported that in the NEXT-11 nations, EG worsened EQ by reducing LCF in the long term, thus making EG unsustainable in these nations.

Caglar et al. (2023) reported that the consequences of EG on LCF is negative in APEC nations during the 1992-2018 period utilizing the CUP-FM analysis. The authors demonstrated that EG in APEC countries is not sustainable. Degirmenci et al. (2025) explored the consequences of EG on LCF in high-income countries. The authors found that, except for Sweden, EG reduced LCF and therefore increased environmental pollution. Pata & Balsalobre-Lorente (2022) stated that EG is the most negatively affecting factor on LCF in Türkiye and that its long-term effects are quite significant. Awosusi & Ozsahin (2026) showed that EG reduces LCF using quantile analyses in Japan. Agila et al. (2022), using quantile regression analysis, stated that the consequence of EG on LCF is negative, but this finding is more pronounced at lower quantiles. The authors stated that EG in South Korea is not sustainable.

Conversely, Durmus et al. (2025), using the Panel Quantile approach in NIC countries, found that EG improved LCF, thus contributing to environmental protection, but EG above a certain threshold reduced EQ. Wang & Xu (2025) showed that EG increased LCF in all quantiles in E7 countries during the period 2000-2022. The authors stated that EG is sustainable in E7 countries and is separated from environmental pollution.

Wang (2024a) determined that in the long term, EG reduces LCF in BRICS countries, but EG improves EQ after a certain threshold. The authors found that higher growth increases LCF by increasing environmental awareness and accelerating investment in green technologies. Güloğlu et al. (2023) argued that LCC suggests a U-shaped linkage between EG and LCF. This hypothesis argues that nations initially increase their demand for fossil fuel energy in order to raise their income levels, and this leads to environmental degradation. Technological development, which increases after a certain turning point in per capita income, makes individuals more reactive to the environment, thereby reducing EF and improving EQ (Güloğlu et al., 2023).

Using the QMG approach for 26 OECD nations, Güloğlu et al. (2023) verified the LCC hypothesis for the 1980-2018 period. For rapidly growing economies, Wu et al. (2024) reported that the LCC hypothesis is valid. Wang et al. (2024a) confirmed the LCC hypothesis in Asian nations. For BRICS nations, Li et al. (2024), Wang et al. (2024b), and Caglar et al. (2024) validated the LCC hypothesis in their studies. On the contrary, Ge & Mehmood (2024) cannot validate the LCC hypothesis using the MMQR method based on the period between 1900 and 2022 in OECD nations.

2.5 Research gap

The literature review showed that the number of studies emphasizing the LCF increases, but the findings of studies addressing the factors affecting the LCF differ from one another. This difference stems from the econometric method used, income level, regional characteristics, institutional quality, and uncertainty in climate policies. Second, although there is a limited number of studies assessing the consequences CPU on the LCF specifically in China, studies considering the moderating effect of FD have been overlooked. Therefore, the alarming pollution levels in China encourage us to conduct further research in China. Few studies on CPU directly scrutinized the linkage between climate action and the EQ. This study explores the direct consequences of CPU on LCF and identifies a moderating effect with FD. Thus, in this period where globalization is seen in almost every country, the consequences of FD and CPU on the EQ can be discussed in detail. In addition, the study has been expanded with energy consumption and EQ, whose effects on EQ have been demonstrated in numerous studies. Hence, this study aims to enrich the literature with current period and current empirical methods and in terms of the relationships investigated. This study, conducted on China, one of the

highest polluting countries, is considered to have global importance in terms of its results.

Existing empirical studies have comprehensively examined the direct impact of FD on EQ, but the results are complex. Furthermore, studies examining the interaction between FD and climate policy uncertainty are quite limited. These research gaps restrict the understanding of the complex mechanisms by which financial and policy factors collectively influence environmental outcomes. This study aims to contribute to the literature by addressing these gaps.

3. Data, model construction and methodology

3.1. Data and model construction

This section introduces models for the relationship between EQ and FD, climate uncertainty index, energy consumption, and EG. Information about the dataset used in the study, which was conducted with the help of Chinese data for the 1990-2021 period, is provided in Table 1.

Table 1. Data Set

Acronym	Variable Definition	Data Source
LCF	Biocapacity/Ecological Footprint (gha per person)	Global Footprint Network (2025) https://www.footprintnetwork.org/
FD	Financial Development Index	International Monetary Fund (2025) https://www.imf.org/en/Data
CPU	Climate Policy Uncertainty Index	Gavrilidis (2021) https://ssrn.com/abstract=3847388
EC	Energy use (kg of oil equivalent per capita)	World Bank (WDI) (2025) https://databank.worldbank.org/source/world-development-indicators
GDP	GDP per capita (constant 2015 US\$)	World Bank (WDI) (2025) https://databank.worldbank.org/source/world-development-indicators
INDEX	FD*CPU	International Monetary Fund (2025), Gavrilidis (2021) https://www.imf.org/en/Data , https://ssrn.com/abstract=3847388

Note: The interaction variable INDEX was generated by multiplying the mean-centered values of FD and CPU. CPU is an index created based on Climate issues. FD indicates how developed financial institutions and financial markets are in terms of depth, reach and efficiency.

The empirical models were established by considering the studies of Ridvan et al. (2024), Çelik & Güriş (2025), Pata (2025) and theoretical expectations. These empirical models are specified in equations (1) and (2).

$LCF_t = \beta_0 + \beta_1 FD_t + \beta_2 CPU_t + \beta_3 EC_t + \beta_4 GDP_t + \varepsilon_{1t}$	(1)
$LCF_t = \beta_5 + \beta_6 FD_t + \beta_7 CPU_t + \beta_8 EC_t + \beta_9 GDP_t + \beta_{10} INDEX_t + \varepsilon_{2t}$	(2)

In equations (1) and (2), t represents the sample period (1990-2021); β_0 and β_5 denote the constant terms of models (1) and (2), respectively. The error terms are symbolized as ε_{1t} and ε_{2t} , respectively. All of the variables are in their natural logarithms. This transformation is considered to prevent issues with respect to the dynamic properties of the series (Shahbaz et al., 2012; Bhattacharya et al., 2016). The parameters $\beta_1, \beta_2, \beta_3, \beta_4$ and $\beta_6, \beta_7, \beta_8, \beta_9, \beta_{10}$ are the elasticity coefficients of the respective variables in models (1) and (2), respectively. In order to identify the aforementioned regulatory effect of FD, an interaction term between FD and climate policy uncertainty was added to the econometric model in this study. This modification to the model allows for understanding whether the impact of climate policy uncertainty on EQ varies at different levels of FD.

3.2. Methodology

In econometric analyses, the problem of spurious regression can arise in analyses using non-stationary series (Ghouse et al., 2024). This can result in biased results in empirical estimates. Therefore, in the empirical methodology, a unit root test is first utilized for the variables. To determine this, the ADF unit root test introduced by Dickey & Fuller (1981) and the PP unit root test proposed by Phillips & Perron (1988) are applied. The null hypothesis of both tests is the identification of a unit root. However, the alternative hypothesis indicates the stability. The ADF test assumes that the residuals are not autocorrelated. The ADF unit root test accounts for prior values of the dependent variable into the model. This demonstrates that the autocorrelation problem can be overcome. The PP test is more robust than the ADF test because it can be applied to time series that contain serial correlation and heterogeneity (Phillips & Perron, 1988). The KPSS test proposed by Kwiatkowski, Phillips, Schmidt, & Shin (1992) demonstrates null hypothesis stationarity, unlike the ADF and PP. In this test, the series constituting the model are stabilized by removing trends with deterministic properties (Göktaş et al., 2018). Structural breaks are not considered in all three traditional tests. This can introduce bias into results in determining the root cause (Becker et al., 2006). Becker et al. (2006) proposed the Fourier KPSS test using Fourier functions to account for sudden and slow changes in a test they designed.

Some degenerate processes have emerged in the ARDL boundary test, which has become popular and widely used due to its many advantages. In the ARDL bounds test, which draws on the F-statistic for the significance of each variable and the t-statistic for the significance of the explanatory variable, some researchers have recently raised concerns about the use of only the F-statistic. Another limitation is that the traditional ARDL test cannot be utilized when the explanatory variable is stationary at level $I(0)$. To address these issues, referred to as degenerate case I and degenerate

case II, McNown et al. (2018) applied a bootstrap approach, and Sam et al. (2019) developed the Augmented ARDL (A-ARDL) approach.

The equations formed by considering Model-1 and Model-2 are given in (3) and (4) respectively.

$\begin{aligned}\Delta \ln LCF_t = & \alpha_0 + \alpha_1 \ln LCF_{t-1} + \alpha_2 \ln FD_{t-1} + \alpha_3 \ln CPU_{t-1} \\ & + \alpha_4 \ln EC_{t-1} + \alpha_5 \ln GDP_{t-1} + \sum_{i=1}^p \beta_{1i} \Delta \ln LCF_{t-i} \\ & + \sum_{i=0}^p \beta_{2i} \Delta \ln FD_{t-i} \\ & + \sum_{i=0}^p \beta_{3i} \Delta \ln CPU_{t-i} + \sum_{i=0}^p \beta_{4i} \Delta \ln EC_{t-i} \\ & + \sum_{i=0}^p \beta_{5i} \Delta \ln GDP_{t-i} + u_t\end{aligned}$	(3)
$\begin{aligned}\Delta \ln LCF_t = & \alpha_0 + \alpha_1 \ln LCF_{t-1} + \alpha_2 \ln FD_{t-1} + \alpha_3 \ln CPU_{t-1} \\ & + \alpha_4 \ln EC_{t-1} + \alpha_5 \ln GDP_{t-1} + \alpha_6 \ln INDEX_{t-1} \\ & + \sum_{i=1}^p \beta_{1i} \Delta \ln LCF_{t-i} + \sum_{i=0}^p \beta_{2i} \Delta \ln FD_{t-i} \\ & + \sum_{i=0}^p \beta_{3i} \Delta \ln CPU_{t-i} + \sum_{i=0}^p \beta_{4i} \Delta \ln EC_{t-i} \\ & + \sum_{i=0}^p \beta_{5i} \Delta \ln GDP_{t-i} + \sum_{i=0}^p \beta_{6i} \Delta \ln INDEX_{t-i} + u_t\end{aligned}$	(4)

In equations (3) and (4), the intercept is represented by a_0 . The difference operator is symbolized by Δ . The delay length is symbolized by p , and the error term by u_t . The long-term coefficients are symbolized by $a_1, \dots, a_6$ while the short-run coefficients are represented by $\beta_1, \dots, \beta_6$.

The ARDL methodology was first introduced by Pesaran et al. (2001). In this test, a cointegration linkage is checked with F and t tests. McNown et al. (2018) and Sam et al. (2019), who developed this approach, proposed a new F test to check the statistical significance of the lags of the independent variables in the model. Thus, they suggested checking the validity of a cointegration linkage with three different tests. Pesaran et al. (2001) added the condition of testing the null hypothesis, as stated in (5) (6), for Model-1 and Model-2 respectively (Mert & Çağlar, 2023).

$$H_0: \beta_2 = \beta_3 = \beta_4 = \beta_5 = (5)$$

$$H_0: \beta_2 = \beta_3 = \beta_4 = \beta_5 = \beta_6 = (6)$$

The hypotheses given in equations (5) and (6) are null hypotheses of the F test added in addition to the traditional ARDL method. The Augmented ARDL (A-ARDL) method assumes that if all three test hypotheses are rejected or if the relevant three test statistics are greater than the upper limit value, a cointegrated relationship is concluded.

The A-ARDL method examines short and long-run dynamics besides cointegration linkage. In this study, in addition to the ARDL methodology,

different cointegration estimators are also utilized to enhance the robustness of the results. In this context, two different long-run coefficient estimators are utilized in this study. The first is the Fully Modified Ordinary Least Squares (FMOLS) estimator. Developed by Hansen & Phillips (1990), this method uses a nonparametric approach to correct for both autocorrelation and endogeneity. FMOLS performs unbiased long-run estimation in cointegrated models. Following this method, the Dynamic Ordinary Least Squares (DOLS) estimator is used as a robustness test. The DOLS estimator proposed by Stock & Watson (1993) adds lagged and forward difference terms to the empirical model. Thus, it performs more robustly than FMOLS in small samples. Furthermore, the DOLS estimator eliminates both autocorrelation and endogeneity problems.

4. Empirical Findings

In this section, we first present the summary statistics and correlation matrices of the series in the models. This information is depicted in Table 2.

Table 2. Descriptive Statistics and Correlation Matrix

Stat.	LCF	FD	CPU	EC	GDP	INDEX
Mean	1.18389 9	0.78721 3	4.53092 7	7.21619 5	8.19170 9	3.74371 4
Median	1.22436 9	0.80052 6	4.48061 4	7.26200 3	8.20757 9	3.50886 3
Max.	0.67625 1	0.39408 1	5.38740 0	7.87639 4	9.34745 3	4.99331 9
Min.	1.61412 5	1.26198 2	4.05291 3	6.60243 5	6.82140 3	3.13209 8
Std. Dev.	0.32994 3	0.26245 6	0.36891 7	0.44503 4	0.78504 9	0.57330 7
Skew.	0.14181 2	0.17855 5	0.90184 7	0.00114 8	0.13755 7	0.76895 4
Kurt.	1.42589 9	1.93183 5	3.09024 7	1.35973 0	1.72619 4	2.42129 2
J-B	3.41098 2 (0.18168 3)	1.69133 7 (0.4292 70)	4.34861 0 (0.1136 87)	3.58731 9 (0.1663 50)	2.26435 9 (0.3223 30)	3.60008 3 (0.1652 92)
Cor. Matrix	LCF	FD	CPU	EC	GDP	INDEX
LCF	1					
FD	0.79171 6	1				
CPU	0.66025 7	0.63878 1	1			
EC	0.79434	0.76207 0	0.70593 3	1		

	1					
GDP	-	0.77972	0.68415	0.78328		
	0.78723	2	6	7	1	
	9					
INDEX	0.17694	-	0.34441	-	-	1
	3	0.09006	9	0.14104	0.09951	
		8		4	3	

Table 2 depicts the summary statistics and correlation matrix for the LCF, FD, CPU, EC, GDP, and INDEX variables in China for the period 1990-2021. All variables in Table 2 exhibit normal distribution at a 5% significance level. An examination of the correlation matrix implies no high correlation.

The literature suggests that high pairwise correlations can be a preliminary finding for some statistical problems (Kutner et al., 2004). However, according to the findings in Table 2, the pairwise correlations are not high. These results raise expectations that multicollinearity can not arise. Therefore, the results obtained from the empirical analyses conducted in this study are strengthened and have higher reliability with lower correlation values. Figure 1 presents the graphs of the variables.

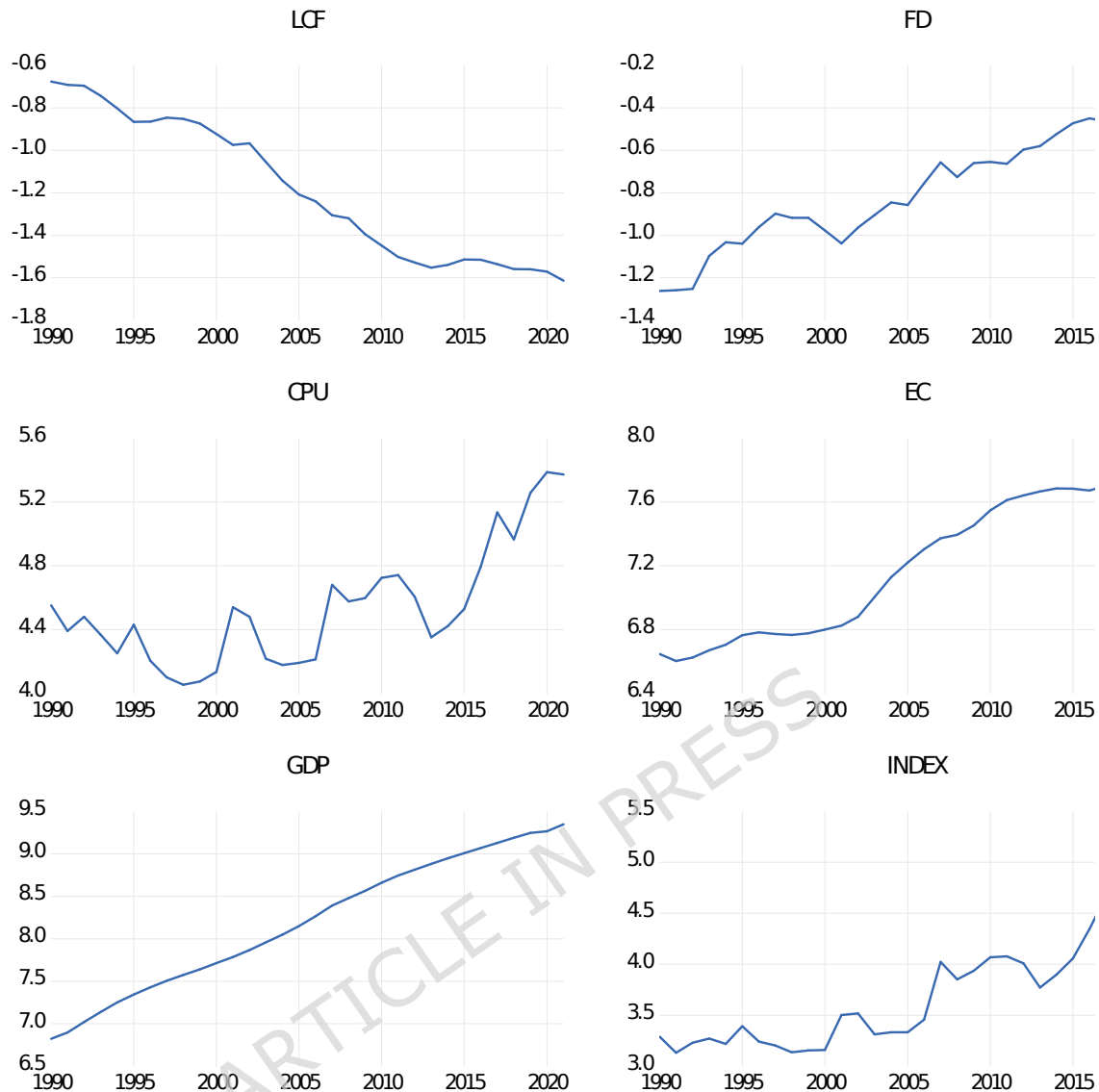


Figure 1. Graphs of Variables

According to the Figure 1, the variables are generally considered to have a trend structure. This preliminary finding indicates that trending results can be prioritized in the empirical methods used.

Detecting unit root processes is important to mitigate spurious regression problems. Table 3 shows the ADF and PP unit root test findings for the variables in models (1) and (2).

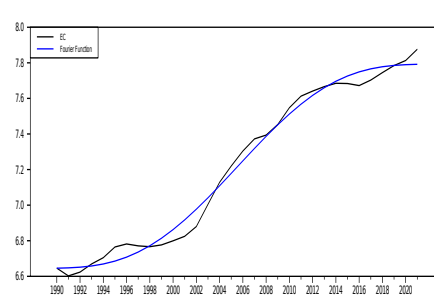
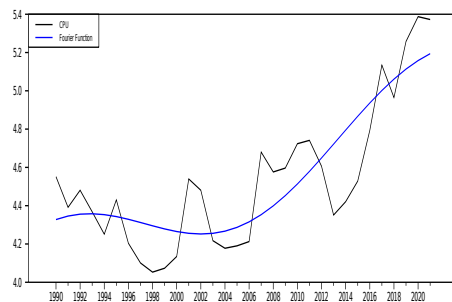
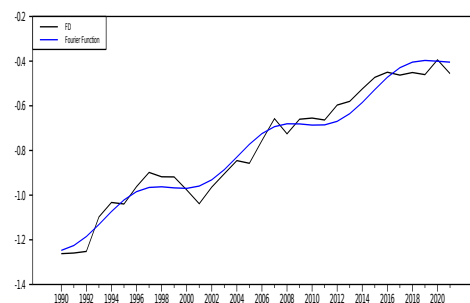
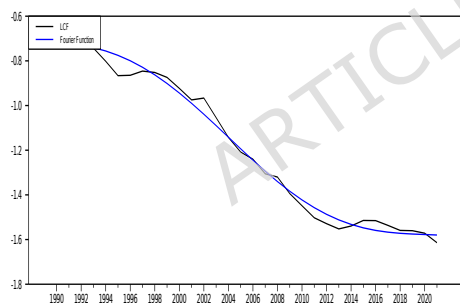
Table 3. Unit Root Test Results (Traditional Methods)

ADF					
LCF	FD	CPU	EC	GDP	INDEX
-1.601686 (0.7683)	-2.450054 (0.3487)	-2.162534 (0.4926)	-2.210823 (0.4669)	-1.820271 (0.9132)	-2.326138 (0.4085)
Δ LCF	Δ FD	Δ CPU	Δ EC	Δ GDP	Δ INDEX
-3.513668	-5.451866	-5.645763	-2.722958	-5.104873	-4.182630

(0.0559)	(0.0006)	(0.0004)	(0.0824)	(0.0015)	(0.0142)
PP					
LCF	FD	CPU	EC	GDP	INDEX
-1.183759 (0.8965)	-2.507473 (0.3225)	-2.157553 (0.4953)	-2.057800 (0.0419)	-0.436550 (0.9814)	-2.341872 (0.4007)
ΔLCF	ΔFD	ΔCPU	ΔEC	ΔGDP	ΔINDEX
-3.460076 (0.0624)	-5.451777 (0.0006)	-6.655044 (0.0000)	-3.048191 (0.0418)	-3.307930 (0.0843)	-7.005002 (0.0000)

Note: In conducting the ADF unit root test, the Schwarz Information Criterion was applied. The Δ symbol indicates the first-difference operator, while the figures in parentheses correspond to the associated probability values

Table 3 depicts the findings of the ADF and PP unit root tests. The test results conducted at the levels reveal that each variable follows a unit root process. Hence, the probability values of the test statistics obtained in the unit root test without difference calculation are greater than 0.10. According to the ADF results, all of the series are stationary at their first difference. According to the PP results, the EC variable is stationary at the level, in contrast, the other variables are stationary at their first difference. Traditional ADF and PP tests fail to account for structural breaks. This can bring about misleading results (Perron, 1989). To overcome this, the Fourier KPSS test, which incorporates structural changes into the model with Fourier functions, can be used. The ability of Fourier functions to mimic variables is illustrated in Figure 2, from left to right, by the variables LCF, FD, CPU, EC, and INDEX.



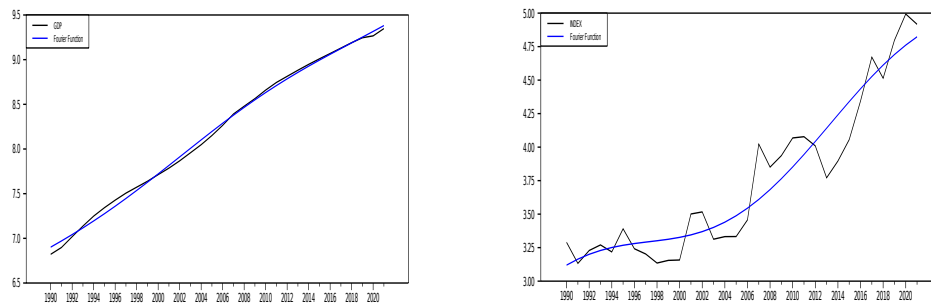


Figure 2. Fourier Functions and The Time Path of Variables

Figure 2 shows that Fourier functions effectively mimic the time course of variables. This results in more consistent results than models that do not include structural breaks (Nazlioglu et al., 2023). Table 4 presents the Fourier KPSS and KPSS test findings.

Table 4. Evidence from Fourier KPSS and KPSS Tests

Variables	Freq.	FKPSS Stat.	F(k)	Min.RSS	KPSS Stat.
LCF	1	0.11028	40.44252	0.03932	0.112195
FD	3	0.09403	9.92699	0.06525	0.059108
CPU	1	0.10921	9.94015	1.20212	0.167644
EC	1	0.11583	36.28817	0.07458	0.096069
GDP	1	0.09553	19.02931	0.04913	0.139518
INDEX	1	0.11569	11.97049	0.97941	0.174614

Note: In KPSS test, the critical values at 10%, 5%, and 1% significance levels are 0.119000; 0.146000; 0.216000, respectively. In Fourier KPSS statistics, the critical values for $k=1$ at 10%, 5%, and 1% significance levels are 0.0471; 0.0546; 0.0716, for $k=3$, 0.1141; 0.1423; 0.2103. In F test statistics, the critical values at 10%, 5%, and 1% significance levels are 4.162; 4.972; 6.873, respectively.

Table 4 presents the Fourier KPSS and traditional KPSS test findings. To use the Fourier KPSS test findings, the significance of $F(k)$ is checked. Here, $F(k)$ is the test statistic for the significance of the trigonometric terms, sine and cosine. All $f(k)$ statistics surpass the critical value of 6.873 specified for a 1% significance level. This guarantees that the FKPSS test statistic can be considered for evaluation. In cases where the $F(k)$ statistic is not significant, the traditional KPSS test outcomes are used for interpretation. According to the Fourier KPSS test findings, the null hypothesis is rejected at the 1% significance level for the LCF, CPU, EC, GDP, and INDEX variables. The relevant variables are non-stationary at the level. For the FD variable, $k=3$ was obtained, and when the relevant critical value was compared with the FKPSS test statistic, we failed to reject the null hypothesis. Hence, FD is stationary at the level. When the findings of Table 3 and Table 4 are evaluated together, all variables have the $I(0)$ and $I(1)$ properties. When the obtained findings are evaluated together, the prerequisite for the A-ARDL cointegration analysis is met. Table 5 presents the results of the A-ARDL cointegration analysis.

Table 5. Empirical Findings of the A-ARDL Cointegration Test

Model-1	k	Stat.	I(1) Crit. Val.
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(0,1,1,1,1)	4	F_{ov} 5.36580*	%1 7.578	%5 5.540	%10 4.624
		t_{pv} -4.05513*	-4.96	-4.36	-4.04
		F_{IV} 3.96367*	7.01	4.71	3.79
Diag. Tests		Stat.	P-Values		
B-G		0.000001	0.9999999		
White		31.286571	0.4518405		
Ramsey RESET		0.2649598	0.6067330		
Jarque-Bera		0.890970	0.640513		
CUSUM			Stable		
CUSUMQ			Stable		
Model-2	k	Stat.	I(1) Crit. Val.		
(0,1,0,0,0)	5	F_{ov} 5.01800*	%1 7.242	%5 5.253	%10 4.412
)		t_{pv} -4.65272**	-5.13	-4.52	-4.21
		F_{IV} 6.27242**	6.66	4.49	3.67
Diag. Tests		Stat.	P-Values		
B-G		6.661338e-015	0.9999999		
White		30.002248	0.6171450		
Ramsey RESET		0.8352266	0.3607655		
Jarque-Bera		1.049835	0.591604		
CUSUM			Stable		
CUSUMQ			Stable		

Note: The relevant significance levels at critical values refer to the upper bound values of I(1). The 10% and 5% significance levels are highlighted with *, ** respectively.

Table 5 presents the A-ARDL findings for Model-1 and Model-2, including constants and trends. Three different test statistics were compared against the critical values proposed by Narayan (2005), Pesaran et al. (2001), and Sam et al. (2019) respectively. When the existence of cointegration is proven, the A-ARDL method determines the presence of cointegration. Thus, more robust findings can be obtained compared to the traditional ARDL method (Özbek et al., 2025). When the calculated statistic surpasses the upper bound value of I(1), the presence of cointegration is confirmed. The empirical findings indicated a cointegrating linkage in both models. Moreover, model diagnostics should also be evaluated. The test findings revealed that there is no serial correlation issue, variance is constant, no model construction errors are detected, and normal distribution validity is achieved. As a result, the existence of a cointegrated relationship is determined. Table 6 depicts the FMOLS results.

Table 6. FMOLS Results

FMOLS (Model-1)				
Variables	Coeff.	Std. Err.	t-Stat.	P-value
FD	-0.254608	0.030789	-8.269342	0.0000
CPU	-0.023632	0.008182	-2.888131	0.0079
EC	-0.488028	0.024785	-19.69020	0.0000

GDP	-0.603555	0.045023	-13.40546	0.0000
C	6.892617	0.269075	25.61597	0.0000
T	0.031113	0.003040	10.23528	0.0000
FMOLS (Model-2)				
Variables	Coeff.	Std. Err.	t-Stat.	P-value
FD	-0.248296	0.030868	-8.043766	0.0000
CPU	-0.019583	0.009460	-2.070025	0.0494
EC	-0.482198	0.025438	-18.95619	0.0000
GDP	-0.593554	0.044109	-13.45643	0.0000
INDEX	0.031015	0.010440	2.970775	0.0063
C	6.792674	0.274998	24.70077	0.0000
T	0.030329	0.002994	10.13003	0.0000

Note: C indicates the constant term and T indicates the trend.

Table 6 presents the estimation results of models (1) and (2). According to the FMOLS results in Model (1), all variables are statistically significant at the 1% level. Hence, the coefficients obtained are either greater than the critical value obtained at 1%, or the probability value of the coefficients is less than 0.01. A 1% upsurge in FD, CPU, EC, and GDP decreases the LCF by 0.25%, 0.02%, 0.49%, and 0.60%, respectively. According to the results in Model (2), the CPU variable is statistically significant at the 5% level, and the other variables are statistically significant at the 1%. A 1% increase in FD, CPU, EC, GDP, and INDEX decreases the LCF by 0.25%, 0.02%, 0.48%, 0.59%, and 0.03%, respectively. In both models, the constant term and the trend are statistically significant. According to these results, the interaction variable consisting of FD and CPU positively affects EQ. The positive coefficient of the interaction term obtained from econometric analyses highlights that FD somewhat mitigates the negative impacts of climate policy uncertainty on EQ. This interaction coefficient, which alleviates environmental problems, shows that systems with increased financial depth protect the environmental sustainability of economies by providing flexibility and resilience to the current situation. This result suggests that advanced financial systems can mitigate the negative environmental externalities of policy uncertainty by opening access to environmentally conscious credit and supporting investments in green technologies. DOLS robustness test was conducted to strengthen the obtained results. The findings obtained are given in Table 7.

Table 7. Robustness Check

DOLS (Model-1)				
Variables	Coeff.	Std. Err.	t-Statistic	P-value
FD	-0.340448	0.140689	-2.419865	0.0323
CPU	-0.117123	0.034717	-3.373651	0.0055
EC	-0.317363	0.214084	-1.482420	0.1663
GDP	-0.948718	0.256902	-3.692923	0.0035
C	8.535697	0.986915	8.648867	0.0000
T	0.050075	0.013259	3.776824	0.0031
DOLS (Model-2)				
Variables	Coeff.	Std. Err.	t-Statistic	P-value
FD	-0.313645	0.154530	-2.029672	0.0820

CPU	-0.159380	0.044094	-3.614530	0.0086
EC	-0.104120	0.148916	-0.699187	0.5070
GDP	-0.523268	0.160602	-3.258171	0.0139
INDEX	0.035183	0.007178	4.901748	0.0017
C	3.964913	1.044362	3.796492	0.0067
T	0.026071	0.008505	3.065505	0.0182

Note: C indicates the constant term and T indicates the trend.

Table 7 depicts the findings for models (1) and (2). The findings for DOLS in Model (1) reveal that all statistics are significant for CPU, GDP, and FD. A 1% upsurge in FD, CPU, and GDP decreases the LCF by 0.25%, 0.02%, 0.49%, and 0.60%, respectively. According to the results in Model (2), the CPU variable is significant at the 5% level, while the other variables are statistically significant at the 1% level. A 1% increase in FD, CPU, EC, GDP, and INDEX decreases the LCF by 0.25%, 0.02%, 0.48%, 0.59%, and 0.03%, respectively. In both models, both the constant term and the trend are significant. The robustness results are also largely consistent with FMOLS. According to the robustness findings, it was seen that the interaction variable consisting of FD and CPU positively affected the EQ. The findings highlight that FD is not only a direct determinant of EQ, but also a critical regulatory factor shaping the effectiveness of climate policies under uncertainty.

5. Discussions and Concluding Remarks

This study explores the direct effects of FD, CPU, EG, and EC on EQ in China, as well as the indirect effects of the interaction between FD and CPU. The findings showed that FD, CPU, and EG in China have a direct negative impact on EQ. This finding can be explained by the fact that FD in China has been insufficient in meeting the financing needs of R&D investments in clean energy and in improving EQ through green finance instruments. This finding supports Ali et al. (2025)'s finding that FD increases EG, thereby increasing energy demand and contributing to natural resources depletion, which in turn reduces LCF. Similarly, the findings are in line with findings by Özkan et al. (2024) in India, Kartal et al. (2023) in Japan, and Das & Sethi (2023) in low- and middle-income countries, which show that FD impairs EQ. Furthermore, it is consistent with the findings of Latif & Faridi (2022), who examined six different dimensions of FD and showed that FD reduced LCF in 48 Asian countries, thereby deteriorating EQ. However, this finding contradicts the results of Adebayo et al. (2025), which reported that FD improves ecological health, and Topaloğlu et al. (2024), which revealed that FD has no effect on environmental degradation. Boussaidi & Hakimi (2025) similarly argue that the direct impact of FD on EQ is negative. However, the authors stated that the combined moderating effect of FD and institutional quality has an enhancing effect on EQ. Durmuş et al. (2025), who conducted research on Türkiye, also presented results indicating that FD reduces EQ. Some studies, however, argued that the direct impact of FD on EQ is positive,

thus differing from the findings of this study. In this context, Yıldırım et al. (2025) emphasized the importance of FD in ensuring long-term, high-quality environmental sustainability. This result obtained in EU nations shows that the findings differ according to countries and country groups.

The finding that CPU negatively affect ecological balance can be explained by the fact that climate change can reduce EQ through air, water, and soil degradation. Accordingly, as with other uncertainties, CPU also negatively affect policies aimed at protecting environmental health. This result is in line with the study of Sun et al. (2023), which reported that climate change risks reduce the effectiveness of policies aimed at preventing environmental degradation, but it contradicts the results of Liu et al. (2023), which showed that CPU reduce CO₂. Climate policy uncertainty can have negative consequences for EQ. FD can have significant positive effects in mitigating these impacts. Several mechanisms exist in this regard. Firstly, FD encourages firms and households to invest in environmentally friendly ventures despite the climate crisis (Pankratz et al., 2023). Ensuring financial deepening during periods of climate uncertainty encourages relevant economic units to engage in long-term green investments. Secondly, enhanced FD can reduce investment deferral despite climate policy uncertainty. This reduction is based on increased risk sharing and capital allocation efficiency. The diversification of financial instruments provided by developed financial markets can be considered a third mechanism. In this context, economic units can access sustainable financing such as low-cost loans that prioritize EQ (Xu et al., 2022; Lu, 2025). Thus, it is considered that the contribution of production processes to EQ can be increased through these mechanisms. In this context, FD acts as both an institutional and economic buffer by minimizing the negative externalities of CPU. When considered more broadly, it is of great importance to address the problems that can arise from increased climate policy uncertainty through various policy flexibilities. This mechanism can be achieved through FD, as highlighted in this study. On the other hand, as Hussain et al. (2026) stated, the negative impacts of climate vulnerability on sustainable development can be significantly reduced through digitalization. Therefore, it is considered important to focus on inclusive impacts in addition to direct impacts. Therefore, the effects of climate and FD on EQ can vary depending on the country, period, and empirical methods. The pollution levels of countries are particularly important in this relationship. In this context, considering China as a high-pollution country, the connections between financial policies, climate policies, and environmental problems need to be carefully examined.

The negative consequences of EG on EQ indicate that the EG process leads to environmental problems. This result can be explained by increased resource consumption. This mechanism occurs because increased economic activity reduces biocapacity, thereby increasing environmental

stress. These findings are in line with the literature (Duran & Saqib, 2024; Islam et al., 2024; Raihan et al., 2023). The results contradict the studies by Safdar et al. (2019), Adejumo (2020), Yang et al. (2022), and Shafik (1994), which argue that increased growth improves EQ. In that study, the authors emphasize that a positive relationship exists between EG and energy efficiency, potentially reducing environmental degradation.

FD and CPU have negative direct effects on EQ. In other words, increased diversity of financial instruments and climate uncertainty have an effect that increases environmental stress. However, the moderating effects of these variables are positive. More explicitly, the environmental pressure-increasing effect of CPUs can be overcome by the depth provided by FD. These findings, specific to the Chinese economy, necessitate prioritizing financial developments that prioritize EQ. Practices that prioritize EQ are crucial both in overcoming uncertainty and in the effective use of financial instruments. This positive impact can theoretically be realized through three different channels: investment and innovation, and risk-sharing and resilience. When these channels are considered generally, it shows that financial systems can increase green investments. Deepening FD supports environmentally friendly energy projects and R&D activities. Thus, the investment contraction caused by CPU can be limited. This allows for the diversification of risks. With the deepening of financial markets and the increasing diversity of financial instruments, funds can be shifted to areas that ensure resource efficiency (Subhani et al., 2025; Dong et al., 2024). While there are studies in the literature that separately present the environmental impacts of FD and CPU, this study highlights the importance of the interactive effect. Therefore, FD stands out as an important tool for minimizing or even turning the impacts of CPU on EQ into a positive one. Consequently, the results emphasize the importance of supporting sustainable development policies not only with regulatory frameworks but also with strengthening the financial market infrastructure. Thus, it is considered that FD can be a complementary policy tool in managing the environmental costs of CPU.

Findings showing that fossil fuel consumption worsens EQ can be explained by the fact that a large portion of energy production in China is coal-based (Climate Action Tracker, 2025). According to this, the low share of renewable energy sources in total energy production in China can contribute to the deterioration of EQ. This finding supports the findings of Achuo et al. (2022) and Jamel & Derbali (2016) that fossil fuel-based energy consumption increases environmental degradation but contradicts the findings of Ozcan et al. (2020) that energy consumption improves environmental health.

The findings have significant implications for China. Firstly, the period between 1990 and 2021, characterized by structural transformations and the implementation of new market-based policies, clearly demonstrates

that FD alone can not be a factor in mitigating environmental degradation in China. In particular, it has been observed that financing sources generated by early financial deepening were directed towards carbon-intensive sectors, leading to increased environmental costs. Therefore, shifting China's financial system, especially through state-owned banks and capital markets, towards environmentally friendly and green investments is of critical importance. The diversity and scope of financial instruments such as green finance, environmentally conscious credit mechanisms, and sustainable finance taxonomies need to be expanded. Maintaining stability in China's climate policies, eliminating regional implementation differences, and providing a long-term and predictable policy framework instead of short-term planning are crucial. Considering China's recent steps towards net-zero emissions within the framework of environmentally friendly financial practices and sustainable development goals, it is assessed that strengthening institutional capacity and increasing political stability can make the balancing role of FD in the face of uncertainty more effective. This perspective offers important policy implications not only for China, an emerging market economy, but also for other emerging market economies undergoing similar structural transformation processes.

The results obtained in this study gain profound meaning within the context of structural transformations implemented in China during the globalization process. According to the findings, the increase in environmental stress along with financial depth becomes particularly evident during China's early industrialization and export-oriented growth periods. The FD that emerged during this structural transformation period in China financed the high use of fossil fuel-based energy in production, heavy industry, and various infrastructure investments. These developments led to a decline in EQ. This situation, specific to China, proves the existence of the scale effect, a concept found in the environmental economics literature. Another strong inference stems from the negative impact of climate policy uncertainty on EQ. This result is closely related to the environmental policies implemented by China during the relevant sample period. Until the early 2000s, when China achieved high EG rates, it pursued a relatively flexible environmental policy. Furthermore, the heterogeneous structure of local government practices in China regionally prevented the formation of a common strategy towards environmental issues. It is assessed that these developments led to the partial exclusion of green investments in China. In China, these mechanisms have led firms to adopt a "wait and see" strategy at the micro level. As a result, investments in green technologies have been postponed. This finding highlights the significant role of policy credibility in investment decisions in developing economies. However, the positive interaction between FD and climate policy uncertainty has led to a very serious policy implication specifically for China. This result becomes significant in China after 2010. During this period, the achievement of financial depth, the

existence of developed capital markets, and the increase in green financing instruments increased China's investment capacity. Moreover, these investments have begun to become more environmentally friendly. This result emphasizes that FD is not only a factor that increases environmental pressures, but can also function as a risk-sharing and adaptation mechanism that absorbs uncertainty under appropriate conditions. Therefore, this study shows that the environmental impacts of FD during the sample period, which includes China's significant structural transformation process, are not linear and unidirectional. In large and transforming emerging market economies like China, it has been shown that this relationship can vary depending on the sample period, current policy practices, and institutional capacity. The findings regarding China strongly support the necessity of policy approaches that are structured with context-sensitive interaction variables on environmental sustainability. In this context, revealing interaction effects and drawing conclusions in this and similar empirical and theoretical studies is of serious importance.

6. Conclusion, Limitations, and Future Research Directions

This study, focusing on the drivers of environmental sustainability, concentrates on the factors that determine EQ. In this context, the direct effects of FD, EG, and CPU on achieving environmental sustainability are examined. On the other hand, the moderating effect of FD and CPU on EQ is also among the main focuses of the study. Empirical results obtained in the study showed that the direct effect on EQ, the indicator of EQ, is negative. On the other hand, the indirect effects of FD and CPU on EQ were found to be positive. These results open up some policy implications. Firstly, when the negative effect of the FD on EQ is evaluated, it is understood that credit mechanisms are not effective. This result, and the assessment that FD is ineffective, can lead to incorrect policies. This finding can be interpreted as FD being ineffective in achieving environmental sustainability in China. In this context, it is considered crucial to increase green finance instruments and financing sources that prioritize EQ in developments occurring in financial markets.

The fact that the direct impact of CPU on EQ is negative reveals the negative consequences of climate problems on the environment. This result indicates that policymakers in China need to take swift steps to eliminate uncertainty to minimize the consequences of the climate crisis. In this context, establishing a clear roadmap for sustainable development goals and implementing transparent practices is considered important. While the direct impacts of CPU and FD on EQ are negative, the study found a positive moderating effect. In other words, when CPU and FD are considered together, they improve EQ. This result shows that when climate policy uncertainty and FD are considered together, they increase EQ. In China, environmental policies should take this result into account and consider that positive results can be obtained even in an environment of climate

uncertainty when financial resources are used to ensure environmental sustainability. Therefore, the moderating effect is considered a crucial policy implication for reducing the negative consequences of CPU and FD on EQ and for designing these impacts together.

The negative impact of EG on EQ indicates that EG in China is not sustainable. Although rapid EG in China forms the basis of China's trade policy, it is essential that policymakers develop new policies that consider the potential costs of EG to the Chinese economy. For this reason, considering the potential role firms play in improving EQ firms (Lestari & Soewarno, 2024) in China should be encouraged to adopt clean production technologies. In China, it is considered that the state can provide significant incentive mechanisms and funds to encourage green production in order to improve EQ. Initiating green codes or green belts to promote environmentally conscious production, and implementing certain tax incentives, are seen as important steps. These measures are expected to make significant contributions to environmental sustainability in China.

The conclusion that energy utilization has a disruptive effect on environmental sustainability can be explained by China's energy consumption portfolio. In this context, it is considered that China needs green energy sources, and improving EQ and reducing environmental degradation are important steps. However, considering China's high population, policies that reduce energy consumption per unit of output are crucial for both mitigating environmental problems and achieving sustainable development goals. In this context, steps should be taken to increase education aimed at reducing per capita energy consumption, encourage public service announcements, and increase R&D efforts for the development of new technologies. Increasing incentives for prioritizing green energy sources is also considered another important policy conclusion.

This study has made significant contributions to the relevant literature in terms of its findings. Firstly, it comprehensively examines China's financial, environmental, and sustainability policies during the 1990-2021 period, a time of significant structural transformations and almost entirely encompassing the era of globalization. This approach, specifically designed for China, delves deeply into the finance-environment-sustainability relationships during China's transition to a market-based structure, yielding unique results specific to China, a globally significant country, and offering original contributions to the literature. A second contribution is the inclusion of climate policies within the finance-environment-sustainability framework. Within the model discussed in this study, a dimension that has been limitedly explored in the literature is systematically presented. This perspective addresses not only the policy level but also the environmental consequences of predictability, expanding the theoretical scope. The inclusion of an interaction-based analytical perspective constitutes the

third contribution. The majority of existing studies in the literature have directly addressed the environmental impacts of FD. However, this study explicitly modeled the interaction between FD and policy uncertainty and offered a perspective that emphasized the conditional and regulatory role of financial systems. This contribution went beyond the classical viewpoint in the finance-environment debate, providing a more holistic and realistic perspective. The fourth contribution was the use of environmental indicators. By using the LCF, the study moved beyond the most frequently used carbon dioxide usage framework in the literature. Thus, instead of the traditional approach, an environmental sustainability indicator that considers both supply and demand aspects was used, allowing for assessments from a broader and more systemic perspective.

Like other studies, this study has limitations from different perspectives. First, this research takes a holistic approach, considering the economic, social, and political determinants of EQ together, but while considering EG as an indicator of economic progress, it excludes other economic indicators. Therefore, it is recommended that other dimensions of economic progress be considered in future studies. Secondly, this study considers CPU as an indicator of EQ but ignores other policy uncertainties. Considering uncertainties other than climate policy in future work could help to highlight the impacts of different dimensions of uncertainty. Thirdly, this study focuses on FD as a financial determinant of EQ but excludes other financial indicators. Future research can explore the effects of different financial determinants on EQ. Finally, due to data constraints, this study focused on the period 1990-2021. In future research, considering data from recent years along with updates to the database may enable the attainment of comparable results. Additionally, China, considered in this study, is considered a high-polluting country. However, a panel data set can be created by adding other high-polluting countries (G-7, the US, etc.) to China. New-generation panel data techniques, which provide more effective results compared to time-series techniques under data constraints, can be used. This allows policy recommendations regarding EQ at a global scale to be presented, and empirical models can be constructed with a larger data set by overcoming data constraints.

DECLARATIONS

Ethical approval

Ethical approval was not required for this study as it does not involve human participants or animals.

Consent to participate

Not applicable.

Consent to publish

Not applicable.

Availability of data and materials

The datasets used and/or analyzed in the current study are available from the corresponding author upon reasonable request.

Competing interests

The authors declare that they have no competing interests.

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Non-financial competing interest statement

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Authors' Contributions

S.Ö. and İ.B.: Model construction and derivation, Writing-Original draft preparation, **S.Ö., S.Ş. and İ.B.:** Validation, Writing-Review, Editing & Supervision. **S.Ş. and İ.B.:** Writing-Original draft preparation, Validation, Writing-Review, Editing & Supervision. All authors have read and approved the final manuscript.

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